

Question 1

Part-A: Generation of a Symmetric Key

- To generate a random key, I used the Random module available in PyCryptodome Library.
- For this, I generated a 16-byte key.
- Below is the result for the following, with the key printed in bytes and in integer format:-

```
Generated Key in bytes is: b'\xd3>\x87Pa\xf8\xfc\xdap\xbf\x1c\xb5p\x1d\xd4'  
Key in Int is: 280790094921824980209637574479497403860
```

Part-B: Generation of Bob's Asymmetric keys

- First, I took the input of the prime numbers p and q, and here I have assumed that these will be of at least 1023 digits, also to obtain such numbers, I first generated these using scipy library and then used them directly by pasting these numbers in the input.
- These prime numbers are used:- (Font size reduced)

p:

```
9837007905265682498229031851668849030967595739738361060931680715630057810032410561534213108535913611469484677286906123436909660914675391163419619873813426  
4422387269274116276845568451571251711439249902179319057400184183888288349039676633998020584214803751367607312871042164997875046163430290966918692156771875  
28451683817015104696823678814942321933797399062639936344197501430993012050547158990644841245020735528057766186703775935374608980005557447325248259290601  
7890381235873162501450503262917795596875713544997613489933709239549648263921381485467597440202115772247271289079475649683331605571138948280101131007696436  
79384200843735186528563101693380060050548402442976140300866235035221322726193393705750496463749614273779970719082475117981634039133207948210321200127814732  
1133277081586109777675527121212474495904605628571653695227180236802069131734369109836522681493581900790609494684330991278804523761763282428100966612862160  
95180789039378025610923138703085340973655314630680581827703886054023554355757514445774284479
```

q:

```
12321718473996840358430331974111306103006548316409688310510235714359649923222932780412153598085268115245149179567795667674231313948295877696361577262133617  
43786923006793977556113714826219982628940941063860603530510006234787692448226262411371614432956404321331827415977691521164373203105950979314136591295120785  
03058011417023131530401715488790772858817590069543593429419374670918470117167320328353002637896669524179247243119320447730874365739632467750893251810735792  
414497758223639631723022605424454860252564377436936054995673066473139827195343444023513192747056272057066639683648282173839853705084508276996610740237467  
2570719900524217058692067363059947601263472230088431242778898391199013724812700204357329569667572250991858776586722935032438746693451941081164412398031019  
4399606049349934179149254585444777369555431394742380677013331264790176960377050409685115799252240727009977024902266791525185493334231605709620183211476316  
31921216206229836088177429011729846620969305787898843185853616435014005735959654310926686449
```

- Following is the screenshot for the same:-

```
Enter the prime numbers: p and q for part b:-----  
Enter the value of p: 98370079052656824982290318516688490309675957397383610609316807156300578100324105615342131085359136114694846772869061234369096609146753911634196198738134264422387269274116276845568451571251  
7114392499021793190574001841838882883490396766339980205842148037513676073128710421649978750461634302909669  
186921567718752845168381701510466968236788149423219337973990626399363441975014309930120505471589906448412  
45020735528057766186703775935746089800055744753252482592906017890381235873162501450503262917795596875713  
54499761348993370923954964826392138148546759744020211577224727128907947564968333160557113894828010113100  
7696436793842008437351865285631016933800600505484024429761403008662350352213227261933937057504964637496142  
7377997071908247511798163403913320794821032120012781473211332770815861097776755271212124764495904605628571  
6536952271802368020691317343691098365226814935819007906094946843309912788045237617632824281009666128621609  
5180789039378025610923138703085340973655314630680581827703886054023554355757514445774284479  
Enter the value of q: 123217184739968403584303319741113061030065483164096883105102357143596499232229327804  
12153598085268115245149179567795667674231313948295877696361577262136174378692300679377556111371482621998  
2628948041063860603530510006234787692448226262411371614432956404321331827415977691521164373203105950979314  
136591295120785093251810735792415748879077285881759006954359342941937467091847011716723022835300263  
789666952417924724311932044773087436573963246775089325181073579241447758223639631723022605424454860252564  
37436936054995673066473139827195343444023513192747056272057906663968364828217383985370508450827699661074  
023746725707199005242170586920673630599476012634722300884312427788983911990137248127002043573295696675722  
5099185877658672293503243874669345194108116441239803101943996606049349934179149254585444777369555431394742  
3806770133312647901769603770504096851157992522407270099770249022667915251854933342316057096201832114763163  
1921216206229836088177429011729846620969305787898843185853616435014005735959654310926686449
```

- After accepting the input, I implemented the steps used in the RSA algorithm to generate a public and private key. As these numbers are very large, I used the GMP library wrapper available for Python: gmpy2.
- Then, I calculated the “N” by simply multiplying the p&q. And then, I calculated the Euler totient function $(\phi(n)) = (p-1)*(q-1)$.
- After this, I took the Public exponent as a constant (as announced on the gc and assigned 65537 to it.
- Next, I calculated the d:-

$$d \equiv (e^{-1}) \bmod \phi(n)$$
- Now the public keys would be (n,e), and the private key would be (n,d).
- Below is the result:-

```
Public Key is:--
N: 121208842035165120497548483075315882273226603388110709242781508226214506464633472046372929499271016797
5609951355403028195920620958909152093896924512177850130042790732700077535709924370727255682934881829254503
7935972814426453007377660144860868815789117989345356367017891338681545280674299457556187045326591721335760
5315130639246016866244697155718281936052145077124496859693386250093741415967010995205509836923805943890411
1338560276523141350955614503701328150418927790960616967376633251030880077436456717403165868776991042913656
1175567194902179414846778118088741105744809939047622157736162951297565078136108794497871989800127647269624
9239026633177180805942255661108063131265547615322162538806628051907739710425676542843866946975959710120270
4219213001258486531626461636478507498487119467936048740048373894915567918607205981160494421301696611208759
9715680362339515231077303268518493255528660412825830558808740020220181516764232799525535622966820293867448
0528921330832421006976441383382177047251408844398626899447641770390045916028959055720479413903781665849856
0232511035646303326445640098063376832788504696912480758030163212158364316726301193998681908536688760941198
8583626504867233757472423565490010479854428457636312300282472412950284428348327710037911848006922872055210
8848066386762982490726241334051535709495928292141183364316880927990564480205328004524132492898372554185999
8171086565492818230215648008141018495278018957782872905517843756403219979388484965592638924535033211732393
7473622001461648674298957983836235625875536329886907798229405905656534108705407077193341826245032250036259
6175141016679873545786406862168880220809805080145264644281116594951952194258303728829615178673989001876633
6496968026854187226238788230974915096109939290675287907723818761990817531686220331000824584869598651007158
4420893551467301727507768671269301160965025778718628534472201083900698484916739702736285669133454184399323
5162524265519973149650962763476502140183168710731951572933017347068448148778384329869008819042649116648718
202708240319050583999205292239703071
E: 65537
```

```

Private Key is:--
N: 121208842035165120497548483075315882273226603388110709242781508226214506464633472046372929499271016797
5609951355403028195920620958909152093896924512177850130042790732700077535709924370727255682934881829254503
7935972814426453007377660144860868815789117989345356367017891338681545280674299457556187045326591721335760
5315130639246016866244697155718281936052145077124496859693386250093741415967010995205509836923805943890411
1338560276523141350955614503701328150418927790960616967376633251030880077436456717403165868776991042913656
1175567194902179414846778118088741105744809939047622157736162951297565078136108794497871989800127647269624
923902663317718080594225566110806313126554761532216253880628051907739710425676542843866946975959710120270
4219213001258486531626461636478507498487119467936048740048373894915567918607205981160494421301696611208759
9715680362339515231077303268518493255528660412825830558808740020220181516764232799525535622966820293867448
0528921330832421006976441383382177047251408844398626899447641770390045916028959055720479413903781665849856
023251103564630326445640098063376832788504696912480758030163212158364316726301193998681908536688760941198
8583626504867233757472423565490010479854428457636312300282472412950284428348327710037911848006922872055210
8848066386762982490726241334051535709495928292141183364316880927990564480205328004524132492898372554185999
8171086565492818230215648008141018495278018957782872905517843756403219979388484965592638924535033211732393
7473622001461648674298957983836235625875536329886907798229405905656534108705407077193341826245032250036259
6175141016679873545786406862168880220809805080145264644281116594951952194258303728829615178673989001876633
64969680268541872262387882309749150961099392906752879077238187619900817531686220331000824584869598651007158
4420893551467301727507768671269301160965025778718628534472201083900698484916739702736285669133454184399323
5162524265519973149650962763476502140183168710731951572933017347068448148778384329869008819042649116648718
202708240319050583999205292239703071
D: 657117377589669458668827929820707889767267531070932984328856521854433589375227316143190897524924733634
0299612685577550361026544191837004119223908314033143891301151217302249850279019926444511408010185298901768
4340924072901090277572709301143883458960184653163939716518253698423074571975809043284148758785694166510455
4197871677435796459660053091028053952413554875907254440225523515850684729131315235615265347193619663156126
7049002338353481575209878895358070888869570534633980934570056519084970192611396183733537286974761089876283
9093626872587154524849478391701324669593221712065247915584459108534490017269015284268335718387862628361043
9211837019818611476742655849240807948771671997863258160092694918028288129060574571195293136099584738479516
6228029652671624649997433807479953525679746508959490323621576283069125554742211042713637607587970245242724
2305586656991161658326836005685653198793552415841701596727731345411454489926788129933057766454235781483807
5162515050196975506640975685730007551282871697130355942044144849773155829101136699654940915886189692745425
5438928691072472522269383639030780389306699367747456534767089410456942960341803357997673119981821496150653
3156890836891274599502494352354351744906554233632369523841359627209713780903935407499800854541672354625644
8056394180037301193520556020132735999558740865926189799664100047865709834309180179974837976519830502529931
5565995579350817632094440038121261056850513256133299709432452414225337821337788510849526267580533094367152
3692548409388417351305091665646537187444682909380149874131458338734931954835666063278311127804810711040933
7260323734157249973114756938738762840221691687870508067152470153390683096115842974189178230484089944986084
1898275643084808287960806173899792598330751524547851877125453179253408691549933673853962013483750975167368
6892292005953502021720947172810022754213561005766116150775935849135006505113840164019438129927008619871281
2826402791095464154686355023342931125541702918610326296910711490305597400388343661844552898030402823986179
72201357142245711442097717437352833

```

Part-C: Encryption of shared symmetric key

- Now, to encrypt the key, I used the following formula:-

$$C = M^e \text{ mod } N$$

Here, C is the cyphertext (output)

M is the key in the integer

N is part of the public key pair

- To compute the cyphertext, I used the powmod function available in the gmpy2.
- Below is the result.

```

The CypherText is:---
6795066925796720367439891640103319276961157100718289454715792793239344248350534720279538530226342484200372
6347616134149220406237477635077704187324892691212779561678441497435278243916263292192750065360212995466370
0296768017965979171563317483913250228311933560280062779284371756886038122453922659742187614455363113117385
3497763161968658247707115103948336148426770547218274301851490389441196573419245322083475862208313337329139
1403590952355933630442353723270796472528717963444589852876580022903155857611699365201203305908065615118848
2073250717386298825743408646801565682764395490472443629954462447910536385809323274808088744195281049377150
0260224069627076854509299923238811411339077322369600833376782475445464743141917252740830808671339520335458
6387528462094257054881793028579234783684248840113949471941197205864073773206108781766385828152325972052989
9773270411173706712076317700041908923372420684180993234698323156763046243329494289152634044457684351555794
2883971592634062960703786581592196909404105698724373465475144153347465308324286146233394589108411465732696
0734338424520934110925349580547710981000523686930779313621180243652798682840267962084873918629947137384995
4787706364868150315447544537395511063246669791741775003548706644975403871662449451784818734700333792061879
6225419752109777797570963537169456393055838171043062612297051249432301266856326330916110998022138458491746
4837506739097892807022715445507474197780668837368306923808328720752571051994841263095385345009132375002407
3417621566772997756055708222768703925633232394340634256178833484733655568268216639125251816033358653864316
2686300167502482486997411072975865966870435519379339952099008355359836239292662592505138455795678500092121
9905313778157987310698456984948617516785706401229737802811732818151228406346101671832966511178376952302712
0412991596159581030292312428117457993914280095314509828932778511955497687702452890084776736819100327718322
3160082413678857104681764177116209896368448613216184874812018700170356620077848912665404836707175795330586
5401230465609315909972156771711

```

Part-D: Decryption of Symmetric key using Bob's Private Key

- To decrypt the message, I used the following formula:-

$$M = C^e \text{ mod } N$$

Here, the M is the Original Message (Key in this case)
C is the ciphertext received by Bob
N is the part of Bob's Private Key (or Public, too).
- Below is the result:-

```

The original message (Key in this case) is:--

```

```

Key in Int: 280790094921824980209637574479497403860

```

```

Key in Bytes: b'\xd3>\x87Pa\xf8\xfc\xdap\xbf\x1c\xb5p\x1d\xd4'

```

Part E: Bob encrypts a message using the shared symmetric key

- Here I directly used the implementation of the Salsa library provided in the PyCryptodome.
- Bob encrypts the message: “THIS IS A MESSAGE FROM BOB TO ALICE WHICH USED SHARED SYMMETRIC KEY” using the shared key received in the previous part.
- Below is the output:-

```
Original Message without encryption is: THIS IS A MESSAGE FROM BOB TO ALICE WHICH USED SHARED SYMMETRIC KEY
The Encrypted Message from bob is: b'\x04yj\xc8v\x96~<b\xbb\xea\xdb*\xce\xfa\x19\xe4\xb3\xab\x89\xce\x19\xbah[:\x14\x06\x13\xd3\xbb\xbc\xf8
\x8a!\xa9\xe6\xbb\xca\xed\xae\x16H}Q\xd2\xd3\xf1\x8cV\x9f\xeb\xcf\x1eB\\6s\xfdUa\xfb\xa5\xf5U\xe1'
Key in Int: 280790094921824980209637574479497403860
```

Part F: Alice uses the shard key to decrypt the message sent by the Bob

- After receiving the encrypted message from Bob, I decrypted it using the Salsa decrypt method, which takes the key and random nonce to get the original message.
- Below is the output:-

```
The Decrypted Message at Alice's End is: THIS IS A MESSAGE FROM BOB TO ALICE WHICH USED SHARED SYMMETRIC KEY
```

References

- RSA Algorithm: <https://brilliant.org/wiki/rsa-encryption/>
- GMP Python Wrapper: <https://gmpy2.readthedocs.io/en/latest/>